BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 92501

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Brass and bronze are the example of:

(A) Compound
(B) Mixture
(C) Element
(D) Substance

Q2. Cl is considered as free radical because it has:

(A) Unpaired Electron
(B) Paired Electron
(C) No Electons
(D) Even no. of electrons

Q3. Gold foil was surrounded by photographic plate or:

(A) ZnO fluorescent screen
(B) ZnS fluorescent screen
(C) $ZNCl_2$ fluorescent screen
(D) Zn Br_2 fluorescent screen

Q4. Element "X" is located in period lll, group ll of periodic table, deduce its electronic configuration.

(A) 2,8,3
(B) 2,8,2
(C) 3,8,3
(D) None of these

Q5. If electronegativity difference is greater than 1.7, then bond will be:

(A) Ionic Bond
(B) Covalent Bond
(C) Coordinate Covalent Bond
(D) Metallic Bond

Q6. Attraction between highly electron deficient hydrogen and highly electronegative forms:

(A) Covalent Bond
(B) Ionic Bond
(C) Hydrogen Bond
(D) Metallic Bond

Q7. Gases molecules have large empty spaces, so they are highly:

(A) Compressible
(B) Diffusible
(C) Compatible
(D) Reliable

Q8. What amount of NaOH is required to prepare0.5M solution in 1 litre:

(A) 80g
(B) 40g
(C) 30g
(D) 20g

Q9. Colloids can:

(A) Scatter light
(B) Not Scatter light
(C) Absorb heat
(D) Evolve heat

Q10. Which of the following is not correct?

(A) Oxygen is neutral to litmus
(B) Oxygen is not combustible
(C) Oxygen supports combustion
(D) Oxygen is a powerful oxidizing agent

Q11. What is the oxidation number of Sulphur in H_2SO_4?

(A) +6
(B) +5
(C) -5
(D) 0

Q12. The elements having lower ionization energies have higher tendency.

(A) To gain electrons
(B) To retain electrons
(C) To lose electrons
(D) To share electrons

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) How many grams are present in 3 moles of Hydrogen and 2 moles of Chlorine?

(ii) What is the maximum capacity of electrons in s,p,d and f sub shell?

(iii) State whether the following elements are metals or non- metals. Beryllium, Carbon, Sulphur, Sodium, Phosphorus, Nitrogen, Francium, Radium

(iv) What is the difference between Octet and duplet rule?

(v) Give difference between double and triple covalent bond with examples?

(vi) At 30°C, a gas sample take up 2.50L.if temperature is raised to 70°C and pressure remains constant, then what will be the new volume of a gas.

(vii) Identify Solute and Solvent in Sugar solution and sea Water.

(viii) What volume of water is required to make 0.3 molar solution from 90cm^3 of 1.5 molar NaCl solution to make 0.5 molar solution.

(ix) Which type of electrode are used in dawn cell give the products formed during electrolysis of Molten NaCl.

(x) Explain chromium electroplating in terms of anodic and cathodic reaction

(xi) a. Which two properties of gold and platinum are used for making jewelry? b. How un-reactivity of platinum makes it useful?

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [3] Calculate the formula mass of NaCl in a.m.u and convert it into grams?.

(b) [4] Draw the atomic structure of: $^{1}_{1}H$, $^{2}_{1}H$, $^{13}_{6}C$ and $^{12}_{6}C$.

QIV.

(a) [3] Name the elements that make Group-Zero in periodic table. What specific term is used for them?

(b) [4] Is Dative bond is an ionic bond or a type of covalent bond.Explain with the help of an example.

QV.

(a) [3] How Boyle's Law is represented graphically and diagrammatically.

(b) [4] What you know about unsaturated solution give suitable example?

QVI.

(a) [3] Sodium acts as oxidizing or reducing agent.Explain with the help of examples.

(b) [4] Write half cell reactions and overall reaction occurring in dry cell.